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Sample Question Set - Unit 4

Course Code: CSE252
Course Title: INTRODUCTION TO ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING
Course Outcome: CO4
Paper: Sample Multiple Choice Questions
Maximum Marks: 40 x 1 = 40

Unit: 4
Unit Title: Unsupervised Learning
Unit Mapped RBT: L3 (Apply)
Total Questions: 40
Suggested Duration: 60 Minutes

Answer Grid (to be filled by the student)

Q. No.	Answer	Q. No.	Answer	Q. No.	Answer	Q. No.	Answer
Q1		Q11		Q21		Q31	
Q2		Q12		Q22		Q32	
Q3		Q13		Q23		Q33	
Q4		Q14		Q24		Q34	
Q5		Q15		Q25		Q35	
Q6		Q16		Q26		Q36	
Q7		Q17		Q27		Q37	
Q8		Q18		Q28		Q38	
Q9		Q19		Q29		Q39	
Q10		Q20		Q30		Q40	

Multiple Choice Questions (40 x 1 = 40 Marks)

Q1. Name the branch of machine learning that works on data carrying no target column.

A) Unsupervised learning
B) Supervised learning
C) Reinforcement learning
D) Transfer learning

Q2. Identify what the value of k specifies in the k-Means algorithm.

A) The number of features to be used
B) The number of clusters to be produced
C) The number of iterations to be run
D) The number of records in each cluster

Q3. State the name of the diagram produced by hierarchical clustering.

A) A scree plot
B) A confusion matrix
C) A dendrogram
D) A scatter matrix

Q4. Name the DBSCAN parameter that fixes how many neighbours a point needs in order to be a core point.

A) eps
B) k
C) The linkage criterion
D) MinPts

Q5. Identify the quantity that the first principal component captures more of than any other direction.

A) Accuracy
B) Variance
C) Correlation with the target
D) The number of records

Q6. State what the confidence of an association rule measures.

A) The share of all transactions holding both items
B) The number of items appearing in the rule
C) The number of passes the algorithm required
D) The share of antecedent transactions that also hold the consequent

Q7. Explain why the result of a clustering cannot be reported as an accuracy figure.

A) There is no known correct grouping to compare it against
B) The clusters are always of unequal sizes
C) The distance measure is chosen after the run
D) The dataset is always too small to score

Q8. Distinguish k-Means from hierarchical clustering by whether the number of clusters must be supplied.

A) Hierarchical needs it beforehand, k-Means does not
B) Both of the two need it supplied beforehand
C) k-Means needs it beforehand, hierarchical does not
D) Neither of the two needs it supplied at all

Q9. Describe what a robot uses unsupervised learning on its sensor data for.

A) Predicting a labelled fault code for each reading
B) Choosing the actions that earn the largest reward
C) Grouping readings into operating modes and spotting unusual ones
D) Converting the readings into written language

Q10. Interpret a Principal Component Analysis in which the first two components explain 45 and 30 per cent of the variance.

A) The two together account for 75 per cent of the spread
B) The two together account for 13.5 per cent
C) The first component is correlated 0.45 with the second
D) The dataset must have had exactly two features

Q11. Predict the effect of one extreme outlier on a k-Means centroid.

A) The centroid is unaffected by the outlier
B) The outlier is automatically discarded
C) The cluster is split into two new clusters
D) The centroid is pulled towards the outlier

Q12. Recognize the task described: readings from a machine are grouped into operating modes with no labels supplied.

A) Classification
B) Clustering
C) Regression
D) Reinforcement learning

Q13. Explain why the features must be standardised before Principal Component Analysis when their units differ.

A) Standardisation removes the outliers from the data
B) Standardisation makes the components correlated
C) Standardisation reduces the number of components
D) Variance depends on the unit, so one column would dominate

Q14. Summarize what association rule mining produces from transaction data.

A) A numeric prediction for each transaction
B) A cluster label for each transaction
C) Rules stating that items tend to appear together
D) A reduced set of features for each item

Q15. Distinguish the support of a rule from its confidence.

A) Confidence is over all transactions, support over the antecedent's
B) Support is over all transactions, confidence over the antecedent's
C) Both of the two are measured over all transactions
D) Both are measured over the antecedent's transactions

Q16. Explain why DBSCAN can label a point as noise while k-Means cannot.

A) DBSCAN grows clusters from dense regions and leaves the rest
B) DBSCAN is given the number of clusters in advance
C) k-Means measures density rather than distance
D) k-Means requires the features to be standardised

Q17. Predict what happens to the inertia of a k-Means result as k is increased.

A) It rises steadily without limit
B) It falls steadily towards zero
C) It remains constant whatever k is
D) It rises and then falls again

Q18. Describe what the elbow method plots.

A) Inertia against the number of clusters
B) Accuracy against the training size
C) Support against the confidence
D) Variance against the component number

Q19. Explain why the labels a clustering assigns to its groups are arbitrary.

A) They are ordered by the size of each group
B) They record the distance to the cluster centre
C) They match the true classes of the records
D) They are names for the groups, not meaningful values

Q20. Distinguish dimensionality reduction from feature selection.

A) Selection builds new combined features, reduction keeps originals
B) Both of the two build new combined features

C) Reduction builds new combined features, selection keeps original ones
D) Both of the two keep only the original features

Q21. Apply k-Means with k set to 2 to the one-dimensional points 2, 4, 20 and 22, and determine the two centroids at convergence.

A) 3 and 21
B) 2 and 22
C) 12 and 12
D) 4 and 20

Q22. Solve the following: a database holds 600 transactions and an itemset appears in 90 of them. Determine its support.

A) 0.09
B) 0.60
C) 0.15
D) 1.5

Q23. Use the following counts to determine the confidence of the rule that if A occurs then B occurs: A appears in 200 transactions and A with B in 150.

A) 0.15
B) 0.75
C) 1.33
D) 0.25

Q24. Apply Principal Component Analysis to a dataset whose components explain 60, 20, 12 and 8 per cent, and determine how many are needed for at least 90 per cent.

A) 2
B) 4
C) 1
D) 3

Q25. Compute the percentage reduction in width when a dataset of 40 features is reduced to 10 components.

A) 25 per cent
B) 10 per cent
C) 75 per cent
D) 40 per cent

Q26. Examine the following and determine the correct explanation: k-Means run twice on the same data with the same k reports two different partitions.

A) The dataset changed between the two runs
B) The distance measure changed between runs
C) The value of k was silently altered
D) The runs started from different random centres

Q27. Compare k-Means with DBSCAN on the shape of cluster each is able to recover.

A) k-Means recovers convex groups, DBSCAN any dense shape
B) DBSCAN recovers convex groups, k-Means any dense shape
C) Both of the two recover only convex groups
D) Both of the two recover clusters of any shape

Q28. Apply the Apriori property to an itemset holding A and B with a support of 0.2, and determine the largest possible support of the itemset holding A, B and C.

A) 0.4
B) 0.2
C) 0.8
D) 1.0

Q29. Use a dendrogram cut that crosses three vertical lines and determine the number of clusters obtained.

A) 1
B) 2
C) 6
D) 3

Q30. Solve the following: a cluster holds the one-dimensional points 6, 10 and 14. Determine its centroid.

A) 6
B) 10
C) 14
D) 30

Q31. Apply DBSCAN with MinPts set to 4 to a point that has 3 neighbours within eps and lies near no core point, and determine its status.

A) A core point
B) A border point
C) A noise point
D) A cluster centre

Q32. Compute the lift of a rule whose support is 0.2, with an antecedent support of 0.4 and a consequent support of 0.4.

A) 1.25
B) 0.5
C) 2.0
D) 0.8

Q33. Examine a clustering of raw sensor data holding temperature in degrees and pressure in pascals, and determine what governs the result.

A) The pressure column, because of its far larger range

B) The temperature column, because it is measured first
C) Both columns contribute equally to the distance
D) Neither column, since the clustering fails to run

Q34. Use Principal Component Analysis on a dataset of three features and determine the maximum number of components it can produce.

A) 2
B) 1
C) 6
D) 3

Q35. Apply hierarchical clustering to 15 records and determine the number of merges before a single cluster remains.

A) 15
B) 14
C) 16
D) 7

Q36. Compare the inertia of a k-Means result at k equal to 1 with the inertia at k equal to 5 on the same data.

A) It is lower at k equal to 1
B) It is identical at both values
C) It is lower at k equal to 5
D) It cannot be computed at k equal to 1

Q37. Examine a rule reported with a confidence of 0.9 and a lift of 1.0, and determine what should be concluded.

A) The rule is among the strongest in the dataset
B) The confidence must have been miscalculated
C) The consequent is common and the rule adds nothing
D) The antecedent almost never occurs at all

Q38. Use the standard procedure to determine the number of clusters for a robot grouping unlabelled terrain readings.

A) Set k to the number of features in the data
B) Try several values of k and compare elbow and silhouette
C) Set k to the number of records in the data
D) Set k to two, since that is always sufficient

Q39. Apply a minimum support of 0.05 to a database of 2,000 transactions and determine the corresponding count.

A) 50
B) 200
C) 400
D) 100

Q40. Relate the position of the bend in an inertia curve to the value of k that should be reported.

A) The bend marks where extra clusters stop paying their way
B) The bend marks the largest usable value of k
C) The bend marks the smallest possible inertia
D) The bend has no bearing on the choice of k

----- End of Sample Question Set -----